The Pulickal Matrix: A Dual-Axis Model of Cognitive and Computational Expense

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Abstract

Difficulty is conventionally treated as a scalar, undifferentiated construct, obscuring the precise mechanisms by which tasks exert demand on human agents. This paper proposes a dual-axis framework—the *Pulickal Matrix*—which disentangles task difficulty into two orthogonal dimensions: *cognitive expense* and *computational expense*. Cognitive expense captures the mental latency of semantic mapping, abstraction, and the integration of novel conceptual schemas. Computational expense models the algorithmic execution cost—the operational effort required to process multi-step procedures once a schema is established. By mapping these dimensions onto a two-dimensional conceptual space, tasks are categorized into four distinct quadrants, providing a diagnostic tool for task analysis. Furthermore, we introduce a mathematical formalization demonstrating that difficulty is not an intrinsic property of a task, but an observer-relative vector transformed by the operator of an agent's domain expertise. Implications for instructional design, interdisciplinary communication, formal systems design, and human-computer interaction are discussed.

1 Introduction

The concept of difficulty is a central parameter in pedagogical theory, cognitive psychology, and human-computer interaction. However, the prevailing operationalization of difficulty often reduces it to a single scalar metric: a task is simply "easy" or "hard." This unidimensional paradigm fails to capture the multi-faceted nature of human effort.

Consider the difference between computing the product of two 10×10 matrices by hand versus grasping the foundational axioms of Homotopy Type Theory. The former requires minimal conceptual adaptation but demands exhaustive, error-prone operational stamina. The latter requires virtually no operational calculation, but demands immense abstract reorientation. Treating both tasks simply as "highly difficult" conflates two entirely different cognitive architectures.

The present work introduces the *Pulickal Matrix*, a formalized, two-dimensional model that separates difficulty into *cognitive expense* and *computational expense*. By orthogonally separating the "cost of understanding" from the "cost of execution," this framework provides a structured, two-dimensional view of task demand, explaining why expertise can substantially alter perceived task difficulty.

2 Situating the Framework: A Literature Review

To establish the utility of the Pulickal Matrix, it is necessary to differentiate it from, and map it onto, existing cognitive architectures, human factors engineering paradigms, and computational frameworks.

2.1 Cognitive Load Theory and Working Memory

Sweller's *Cognitive Load Theory* (CLT) posits that working memory is strictly limited, dividing load into intrinsic, extraneous, and germane categories [7]. While CLT predominantly focuses on the capacity limits of working memory during the acquisition phase of learning, the Pulickal Matrix shifts the focus toward the *architectural nature of operational taxation*. Cognitive Expense ($\mathcal{CE}$) in our model aligns closely with the effort of building and assimilating schemas (germane load). However, the Pulickal Matrix strictly segregates the algorithmic execution (often indiscriminately lumped into intrinsic load within CLT) into its own orthogonal dimension: Computational Expense ($\mathcal{C}\wr\mathcal{E}$). This decoupling allows for a more precise analysis of tasks where the conceptual schema is known, but the execution remains highly taxing.

2.2 The ACT-R Architecture: Declarative vs. Procedural Knowledge

Anderson's ACT-R (Adaptive Control of Thought-Rational) cognitive architecture fundamentally splits human knowledge into declarative memory (static facts and conceptual mappings) and procedural memory (dynamic rules and execution pathways) [1]. The Pulickal Matrix provides a macroscopic, two-dimensional operationalization of this micro-cognitive split. High Cognitive Expense is primarily incurred during the synthesis, retrieval, and structural alignment of novel declarative chunks. In contrast, Computational Expense models the latency and working-memory drag of executing compiled procedural production rules. The matrix illustrates how extensive procedural execution (high $\mathcal{C}\wr\mathcal{E}$) can be intensely demanding even when the declarative structures (low $\mathcal{CE}$) are fully internalized and trivialized.

2.3 Human Factors and the SRK Framework

In the domain of human-computer interaction and systems engineering, Rasmussen's SRK (Skill, Rule, Knowledge) framework models how human agents interact with complex technical systems [5]. The Pulickal Matrix seamlessly overlays onto this established taxonomy, providing it with a dimensional topology. *Skill-based* behaviors—which are highly automated and subconscious—map directly to Quadrant I (Low $\mathcal{CE}$, Low $\mathcal{C}\wr\mathcal{E}$). *Rule-based* behaviors, which require following deterministic, algorithmic "if-then" procedures without requiring deep conceptual formulation, map to the "Grind" tasks of Quadrant II (Low $\mathcal{CE}$, High $\mathcal{C}\wr\mathcal{E}$). Finally, *Knowledge-based* behaviors, required during novel problem-solving, structural diagnostics, and abstract reasoning, map heavily to the upper echelons of Quadrants III and IV, where $\mathcal{CE}$ is heavily taxed.

2.4 Algorithmic Information Theory and Computational Complexity

Perhaps the most robust theoretical analog to the Pulickal Matrix exists within theoretical computer science and algorithmic information theory. In formal algorithmic analysis, a strict conceptual distinction is drawn between Kolmogorov complexity (descriptive complexity) and classical Turing complexity (time/space operational complexity) [4].

Kolmogorov complexity measures the length of the shortest possible program required to mathematically describe a system—representing the effort required to compress, abstract, and specify the logic. This is the direct machine-equivalent of Cognitive Expense ($\mathcal{CE}$). Conversely, Time/Space complexity measures the operational resources (clock cycles and memory states) required to *execute* that program once written, directly mirroring Computational Expense ($\mathcal{C}\wr\mathcal{E}$). The Pulickal framework extends this mathematical dualism from formal, deterministic Turing machines to human cognitive agents, formally proposing that human task difficulty is fundamentally an embodied manifestation of dual-axis algorithmic complexity.

2.5 Relation to Computational Complexity Theory

The Pulickal Matrix bears a conceptual resemblance to classical computational complexity theory. In algorithm analysis, complexity is often decomposed into:

- **Descriptive complexity**: the conceptual difficulty of specifying an algorithm.
- **Time/space complexity**: the operational resources required during execution.

The Pulickal framework extends this distinction from formal machines to human cognitive agents. Cognitive expense parallels the cost of algorithm discovery and representation, while computational expense mirrors execution cost once the algorithm is known.

This analogy suggests that human task difficulty may be understood as a form of embodied computational complexity.

3 Conceptual Foundations

3.1 Cognitive Expense ($\mathcal{CE}$)

Cognitive expense is defined as the mental bandwidth required for semantic mapping, abstraction, and paradigm alignment. It is the cost of moving from a state of syntactic observation to semantic comprehension. High $\mathcal{CE}$ is triggered by:

- High levels of abstraction (e.g., pure mathematics, topology, abstract algebra).
- Unfamiliar representations or novel syntax.
- The requirement to reconcile conflicting prior schemas (e.g., paradigm shifts).

3.2 Computational Expense ($\mathcal{C}\wr\mathcal{E}$)

Computational expense is the operational latency of execution. It is akin to algorithmic time-complexity ($\mathcal{O}(n)$) instantiated in the human brain. High $\mathcal{C}\wr\mathcal{E}$ is triggered by:

- Deeply iterative processes.
- Maintenance of large states in short-term memory during algorithm execution.
- High-precision manual data routing (e.g., large-scale arithmetic, tracking variables in complex "whitebox" deterministic AI models).

4 The Pulickal Matrix and Quadrant Analysis

The intersection of these two orthogonal axes yields the Pulickal Matrix, establishing four domains of task interaction.

4.1 Quadrant I: Low Cognitive, Low Computational Expense

Tasks herein are highly automated or biologically innate. Examples include basic walking, forming native sentences, or single-digit addition. The agent possesses pre-compiled heuristics that bypass both semantic formulation and algorithmic drag.

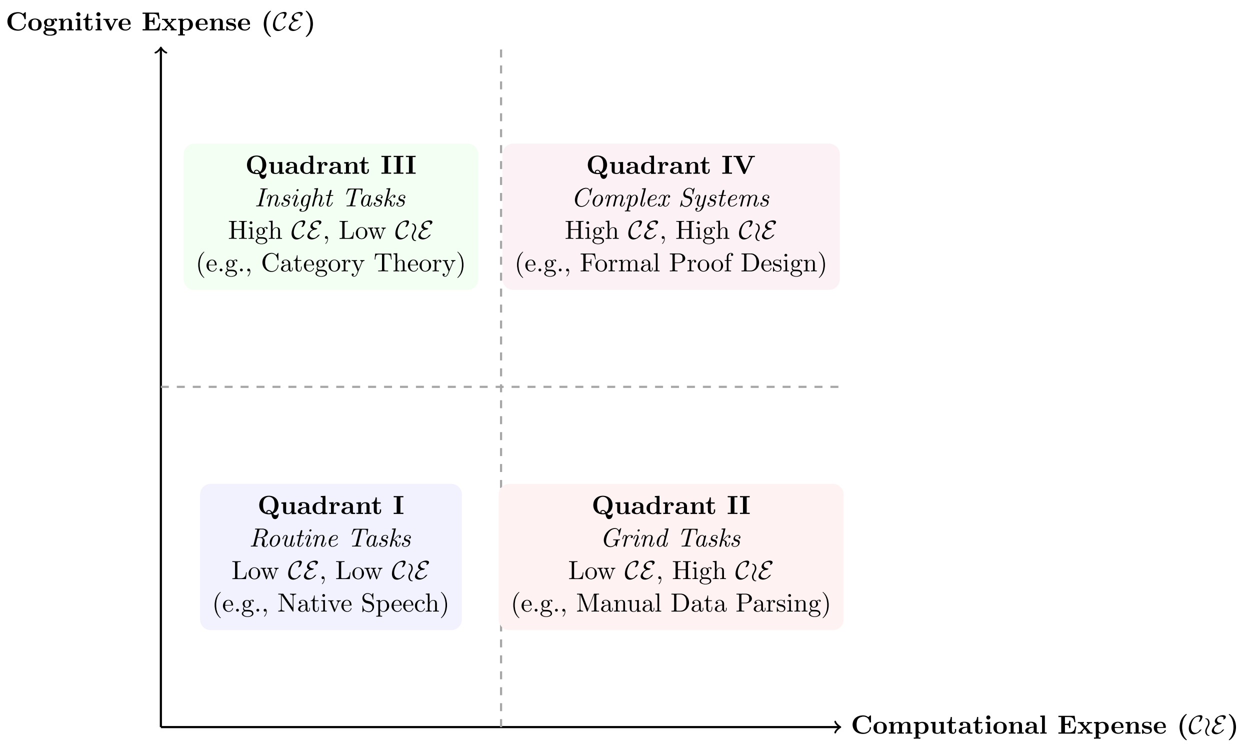

Figure 1: The Pulickal Matrix: A two-dimensional conceptual representation of task difficulty.

4.2 Quadrant II: Low Cognitive, High Computational Expense

These are the ”Grind” tasks. The objective and methodology are transparent, but the execution is grueling. An experienced programmer implementing a well-understood sorting algorithm over a massive dataset exemplifies this quadrant. The logic of the algorithm is fully transparent (low $\mathcal{CE}$), yet tracing edge cases, debugging state transitions, and verifying correctness across numerous iterations impose substantial operational effort (high $\mathcal{C}\wr\mathcal{E}$).

4.3 Quadrant III: High Cognitive, Low Computational Expense

These are the ”Insight” tasks, dominant in pure logic and high abstraction. A reader encountering a foundational theorem in abstract algebra for the first time may face significant conceptual restructuring (high $\mathcal{CE}$), even though the symbolic proof itself is short and procedurally straightforward (low $\mathcal{C}\wr\mathcal{E}$). Once the conceptual schema stabilizes, execution becomes trivial.

4.4 Quadrant IV: High Cognitive, High Computational Expense

These tasks represent the frontier of sustained human effort. They require both the construction of unfamiliar conceptual frameworks and extensive procedural execution. For example, a novice attempting to understand and manually solve a system of nonlinear partial differential equations must first internalize abstract mathematical structures (high $\mathcal{CE}$) while also performing lengthy symbolic manipulations with minimal tolerance for error (high $\mathcal{C}\wr\mathcal{E}$). Similarly, designing and verifying a complex hardware architecture from first principles requires deep conceptual modeling alongside exhaustive logical validation.

5 Mathematical Formalization: Observer-Relativity

A central tenet of the Pulickal Matrix is that difficulty is not an intrinsic scalar of a task T, but a transformation dependent on the agent A.

Let the intrinsic complexity of a task be modeled as a vector $\mathbf{C}_{int} = \begin{bmatrix} c_{\text{cog}} \\ c_{\text{comp}} \end{bmatrix}$.

Let the expertise of an agent A be represented by a diagonal transformation matrix $\mathbf{E}_A$, where elements represent the agent's pre-compiled schemas (e_1) and algorithmic fluency (e_2):[1]

$$\mathbf{E}_A = \begin{bmatrix} e_1 & 0 \\ 0 & e_2 \end{bmatrix}, \quad e_1, e_2 \geq 1$$

The perceived difficulty vector $\mathbf{D}(T, A)$ experienced by the agent is the inverse transformation of the intrinsic complexity by the agent's expertise:

$$\mathbf{D}(T, A) = \mathbf{E}_A^{-1}\mathbf{C}_{int} = \begin{bmatrix} \frac{1}{e_1} & 0 \\ 0 & \frac{1}{e_2} \end{bmatrix} \begin{bmatrix} c_{\text{cog}} \\ c_{\text{comp}} \end{bmatrix} = \begin{bmatrix} \frac{c_{\text{cog}}}{e_1} \\ \frac{c_{\text{comp}}}{e_2} \end{bmatrix} = \begin{bmatrix} \mathcal{C}\mathcal{E} \\ \mathcal{C}\wr\mathcal{E} \end{bmatrix}$$

This formalization provides an illustrative model showing how increasing expertise ($e_1, e_2 \to \infty$) may reduce perceived difficulty, asymptotically shifting tasks toward Quadrant I regardless of intrinsic complexity.

5.1 Scalarization: The Norm of Perceived Difficulty

While the Pulickal Matrix treats difficulty as a two-dimensional vector, practical applications often require a scalar measure of overall perceived difficulty. We therefore introduce a norm over the perceived difficulty vector:

$$D_{\text{total}}(T, A) = \|\mathbf{D}(T, A)\|$$

A natural choice is the Euclidean norm:

$$D_{\text{total}} = \sqrt{\left(\frac{c_{\text{cog}}}{e_1}\right)^2 + \left(\frac{c_{\text{comp}}}{e_2}\right)^2}$$

This scalarization has several important implications:

- Tasks with identical total difficulty may differ radically in their cognitive-computational composition.
- Expertise growth along one dimension may reduce overall difficulty even when the other dimension remains unchanged.
- Pedagogical optimization can be framed as minimizing the norm trajectory over time.

Alternative norms (e.g., Manhattan norm or weighted norms) may be used to model domain-specific sensitivities where cognitive or computational expense is disproportionately taxing.

[1] This diagonal representation assumes independence between cognitive and computational expertise, though future iterations of the model may explore non-zero off-diagonal elements where cognitive mastery accelerates computational speed.

5.2 Temporal Dynamics: Learning as Vector Trajectory

The Pulickal Matrix also provides a natural framework for modeling learning as a time-dependent trajectory in difficulty space.

Let expertise be a function of time:

$$\mathbf{E}_A(t) = \begin{bmatrix} e_1(t) & 0 \\ 0 & e_2(t) \end{bmatrix}$$

As learning progresses, the perceived difficulty vector evolves:

$$\mathbf{D}(T, A, t) = \mathbf{E}_A(t)^{-1}\mathbf{C}_{int}$$

This formulation predicts characteristic learning patterns:

- **Concept-first learning**: rapid decline in cognitive expense followed by gradual computational reduction.
- **Practice-first learning**: early reduction in computational expense with persistent cognitive difficulty.
- **Expert automation**: asymptotic convergence toward Quadrant I.

Thus, learning can be interpreted as a vector field flow toward the origin of the Pulickal Matrix, providing a geometric representation of skill acquisition.

Mathematical Formalization: Trajectory Dynamics

A central tenet of the Pulickal Matrix is that difficulty is not an intrinsic, immutable scalar of a task, but an observer-relative vector that transforms over time as a function of agent expertise.

5.3 Formal Definitions

Definition 1 (Intrinsic Task Complexity)**.** *Let the intrinsic complexity of a task T be a constant vector in a two-dimensional positive orthant $\mathbb{R}^2_+$, defined as $\mathbf{C}_{int} = \begin{bmatrix} c_{cog} \\ c_{comp} \end{bmatrix}$, where $c_{cog}, c_{comp} >$* 0.

Definition 2 (Expertise Operator)**.** *Let $t \in [0, \infty)$ denote time spent in domain-specific learning. The expertise of an agent A is modeled as a time-dependent diagonal transformation matrix $\mathbf{E}_A(t)$, where $e_1(t)$ represents conceptual schemas and $e_2(t)$ represents algorithmic fluency:*

$$\mathbf{E}_A(t) = \begin{bmatrix} e_1(t) & 0 \\ 0 & e_2(t) \end{bmatrix}, \quad e_1(t), e_2(t) \geq 1$$

We assume strictly monotonically increasing expertise functions such that $\frac{de_1}{dt} > 0$ and $\frac{de_2}{dt} > 0$.

The perceived difficulty vector $\mathbf{D}(t)$ experienced by the agent at time t is the inverse transformation of the intrinsic complexity:

$$\mathbf{D}(t) = \mathbf{E}_A(t)^{-1}\mathbf{C}_{int} = \begin{bmatrix} \frac{c_{cog}}{e_1(t)} \\ \frac{c_{comp}}{e_2(t)} \end{bmatrix} = \begin{bmatrix} \mathcal{C}\mathcal{E}(t) \\ \mathcal{C}\wr\mathcal{E}(t) \end{bmatrix}$$

5.4 Dynamics of Learning Trajectories

By treating perceived difficulty as a time-dependent vector, we can formalize the mechanics of learning and adaptation.

Proposition 1 (Monotonicity of Perceived Difficulty)**.** *As an agent's expertise increases monotonically over time, the magnitude of the perceived difficulty vector strictly decreases.*

Proof. Let $||\mathbf{D}(t)||$ be the Euclidean norm (L_2 norm) of the perceived difficulty vector.

$$||\mathbf{D}(t)|| = \sqrt{\left(\frac{c_{cog}}{e_1(t)}\right)^2 + \left(\frac{c_{comp}}{e_2(t)}\right)^2}$$

Given our assumption that $e_1(t)$ and $e_2(t)$ are strictly increasing, their reciprocals $\frac{1}{e_1(t)}$ and $\frac{1}{e_2(t)}$ are strictly decreasing. Because $\mathbf{C}_{int}$ components are positive constants, the sum of their squared scaled values must decrease. Therefore, $\frac{d}{dt}||\mathbf{D}(t)|| < 0$, establishing that the difficulty norm is strictly monotonically decreasing. □

Proposition 2 (Asymptotic Convergence to Routine)**.** *Given unlimited time and ideal learning conditions where* $\lim_{t\to\infty} e_1(t) = \infty$ *and* $\lim_{t\to\infty} e_2(t) = \infty$*, the perceived difficulty of any task* T *asymptotically converges to the origin (Quadrant I).*

Proof. Evaluating the limit of the difficulty vector as $t \to \infty$:

$$\lim_{t\to\infty} \mathbf{D}(t) = \begin{bmatrix} c_{cog} \lim_{t\to\infty} \frac{1}{e_1(t)} \\ c_{comp} \lim_{t\to\infty} \frac{1}{e_2(t)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

Thus, regardless of the initial magnitude of $\mathbf{C}_{int}$, the transformation maps all tasks toward the zero-vector, empirically modeling the transition of complex tasks into routine, automated behaviors. □

5.5 Visualizing Vector Flow

This mathematical formalization yields a dynamic interpretation of the Pulickal Matrix. Instead of static points, tasks exist as trajectories flowing toward the origin (Quadrant I) as expertise scales. The rate and curvature of this flow depend entirely on whether an agent prioritizes conceptual learning (e_1) or operational practice (e_2).

6 Implications and Applications

6.1 Pedagogical Sequencing

Educators often mistake Quadrant III tasks for Quadrant IV tasks. If a student struggles with a topological data analysis concept, giving them more computational exercises (increasing $\mathcal{CIE}$) is highly inefficient. Interventions must specifically target $\mathcal{CE}$ by providing better metaphors and mental models before requiring computational execution.

6.2 Interdisciplinary Asymmetry

Miscommunication between domains (e.g., pure mathematics vs. applied data science) can be mapped via this matrix. A mathematician may view a theorem as Quadrant I (fluent), while an engineer views it as Quadrant III. Conversely, the engineer's highly iterative data-cleaning pipeline may be Quadrant I to them, but Quadrant II to the mathematician who finds the operational drag intolerable.

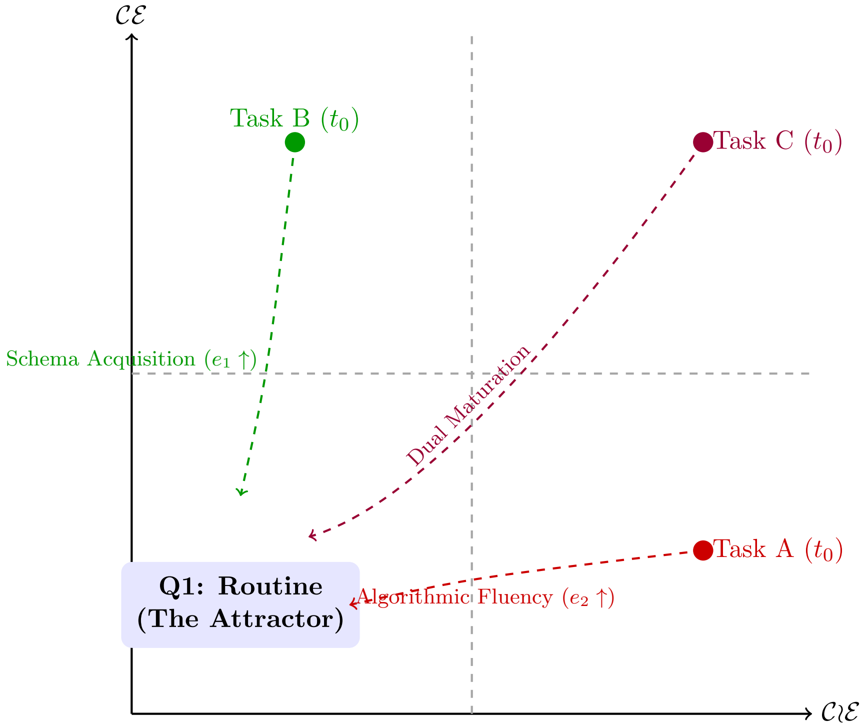

Figure 2: Learning Trajectories in the Pulickal Matrix. As expertise increases, perceived difficulty vectors flow asymptotically toward the origin, shifting tasks from higher-demand quadrants into Quadrant I.

7 Empirical Methodology and Proxy Metrics

To transition the Pulickal Matrix from a conceptual framework to a testable empirical model, it is necessary to establish observable proxy variables for Cognitive Expense ($\mathcal{CE}$) and Computational Expense ($\mathcal{C}\wr\mathcal{E}$). Because internal cognitive states are not directly observable, we propose a temporal and psychometric proxy framework.

7.1 Measurable Proxy Equations

We propose dividing a subject's task engagement into a *Preparation Phase* (schema mapping and strategy formulation) and an *Execution Phase* (algorithmic processing).

Let T_{prep} represent the time elapsed before the first systematic operational action is taken, and let S_{cog} represent the score on the "Mental Demand" subscale of a validated psychometric instrument like the NASA-TLX. The proxy for Cognitive Expense is defined as a linear combination:

$$\mathcal{CE}_{proxy} = \alpha T_{prep} + \beta S_{cog}$$

Let T_{exec} represent the duration of the execution phase (total time-on-task minus T_{prep}), and E_{rt} represent the procedural error rate. The proxy for Computational Expense is defined as:

$$\mathcal{C}\wr\mathcal{E}_{proxy} = \gamma T_{exec} + \delta E_{rt}$$

Where α, β, γ, and δ are empirically derived weighting coefficients dependent on the specific task environment.

7.2 A Proposed Experimental Protocol

To validate the observer-relativity matrix ($\mathbf{D}(T, A)$), we propose a controlled experiment using two distinct cognitive tasks designed to isolate the quadrants of the matrix. The study would involve two cohorts: Novices (no prior domain exposure) and Experts (established domain fluency).

- **Task A (Targeting Quadrant II: Low $\mathcal{CE}$, High $\mathcal{C}\wr\mathcal{E}$):** Manual execution of a deterministic sorting algorithm (e.g., Bubble Sort) on a 100-item physical dataset. The operational rules are conceptually trivial and easily explained, but the execution requires deep working-memory maintenance and exhaustive, error-prone iteration.
- **Task B (Targeting Quadrant III: High $\mathcal{CE}$, Low $\mathcal{C}\wr\mathcal{E}$):** Deciphering and applying a novel artificial grammar rule to translate a single sentence. Formulating the initial structural mapping requires significant abstract conceptualization, but the operational execution (writing the final short sentence) is brief once the rule is understood.

7.3 Concrete Dataset Application (Expected Observations)

By tracking T_{prep} and T_{exec}, we can plot the perceived difficulty vectors of the two cohorts. Table 1 provides a concrete dataset example illustrating the expected outcomes based on the expertise transformation matrix.

Table 1: Projected Experimental Outcomes Demonstrating Observer-Relativity

Cohort	Task	T_{prep} ($\mathcal{CE}$ proxy)	T_{exec} ($\mathcal{C}\wr\mathcal{E}$ proxy)	Error Rate	Mapped Quadrant
Expert	Task A (Sorting)	5s	450s	2%	II (Grind)
Novice	Task A (Sorting)	15s	580s	15%	II $\rightarrow$ IV (Complex)
Expert	Task B (Grammar)	20s	15s	0%	I (Routine)
Novice	Task B (Grammar)	310s	25s	12%	III (Insight)

This experimental framework provides a reproducible methodology for mapping the transformation of intrinsic task complexity into perceived subjective difficulty, laying the groundwork for robust empirical validation of the Pulickal Matrix.

8 Limitations

The Pulickal Matrix is a conceptual framework and should be interpreted as a modeling heuristic rather than a fully empirically validated metric system. First, cognitive expense ($\mathcal{CE}$) and computational expense ($\mathcal{C}\wr\mathcal{E}$) are not directly observable quantities; reliable psychometric instruments for independently measuring these dimensions remain to be developed. Second, the present formulation assumes orthogonality between cognitive and computational expense, though in practice these dimensions may interact dynamically—for example, improved conceptual understanding may reduce operational latency. Third, the diagonal expertise transformation matrix introduced in the formal model simplifies expertise as independent scaling factors; future work may explore non-diagonal transformations capturing cross-dimensional coupling effects. Finally, contextual factors such as fatigue, motivation, and emotional state are not incorporated into the current model, though they may significantly modulate perceived task difficulty. Empirical validation and refinement are necessary to determine the predictive robustness of the framework across domains.

9 Conclusion

The Pulickal Matrix fundamentally reframes task difficulty. By mapping the orthogonal relationship between the mental cost of interpretation and the operational cost of execution, the framework provides a high-resolution diagnostic tool for cognitive performance. Future empirical studies should focus on creating psychometric instruments to independently measure $\mathcal{CE}$ and $\mathcal{C}\wr\mathcal{E}$, paving the way for optimized human learning and more structurally sound human-computer systems.

References

[1] Anderson, J. R. (1983). *The Architecture of Cognition.* Harvard University Press.

[2] Chi, M. T. H., Feltovich, P. J., & Glaser, R. (1981). Categorization and representation of physics problems by experts and novices. *Cognitive Science*, **5**(2), 121–152.

[3] Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.

[4] Li, M., & Vitányi, P. (2008). *An Introduction to Kolmogorov Complexity and Its Applications* (3rd ed.). Springer Science & Business Media.

[5] Rasmussen, J. (1983). Skills, rules, and knowledge; signals, signs, and symbols, and other distinctions in human performance models. *IEEE Transactions on Systems, Man, and Cybernetics*, **SMC-13**(3), 257–266.

[6] Simon, H. A. (1996). *The Sciences of the Artificial* (3rd ed.). MIT Press.

[7] Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, **12**(2), 257–285.